

Contact-Model-Aware Selective Passivation on a 7-DoF Panda

A Simulation Proof of Concept

Abstract

This report asks whether selective passivation remains useful when the planar proof of concept is transferred to a torque-controlled seven-joint robot with spatial contact ports. In simulation, the Panda maintained a 5-N table contact while sliding or wiping. Under end-effector blocking, the dual-ledger filter reduced energy delivered to the human port from 0.263 to 0.049 J. During a forearm interaction it reduced gross output from 0.158 to 0.060 J while maintaining contact and 4.26-mm planar RMSE. A reference governor improved tracking during release without materially changing the protected energy. The result supports selective, discrete port-energy regulation under the stated model and known-wrench assumptions; it is not a claim of general human safety or unconditional robot passivity.

1 Proof-of-concept setup

The simulated system is a hand-free Franka Panda with seven torque-controlled joints, a finite pad with four soft contact elements, and a compliant horizontal table. Control and physics run at 1 kHz. The nominal controller approaches the table, ramps the normal-force target to 5 N, and then follows either a straight path at 0.03 m/s or a bounded two-dimensional wiping path. A scripted spatial wrench is applied at a known end-effector or forearm site (Fig. 1). The known wrench is deliberate: this study isolates the control architecture from force-estimation uncertainty.

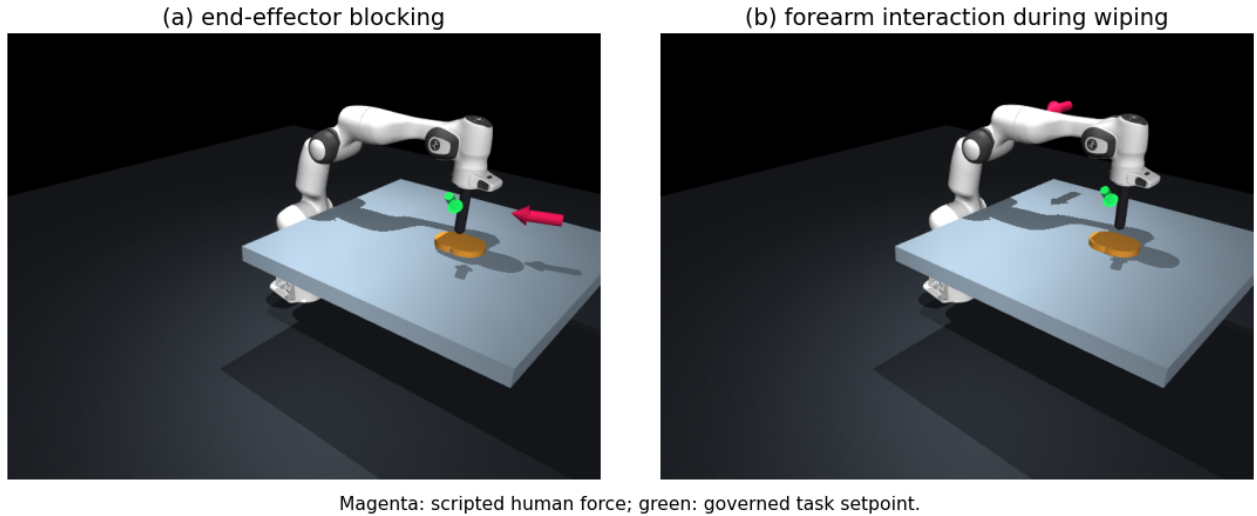


Figure 1: Panda proof-of-concept tasks during the human-interaction phase. The magenta arrow shows the simulator-known human force; the green marker shows the governed task setpoint. These markers are visual only.

Five configurations retain the roles used in the planar report. C3 has no reference governor; C4 is exactly the C3 filter with the governor enabled. Consequently, a C3–C4 comparison attributes changes in task behavior to reference management rather than to a different energy certificate.

Table 1: Controller configurations.

Mode	Accounted port	Protected energy	Governor	Role
C0	none	none	off	nominal baseline
C1	complete external	E_W	off	whole-port baseline
C2	model residual	E_R	off	model-aware baseline
C3	residual and human	E_R, E_H	off	selective filter
C4	residual and human	E_R, E_H	on	governed C3

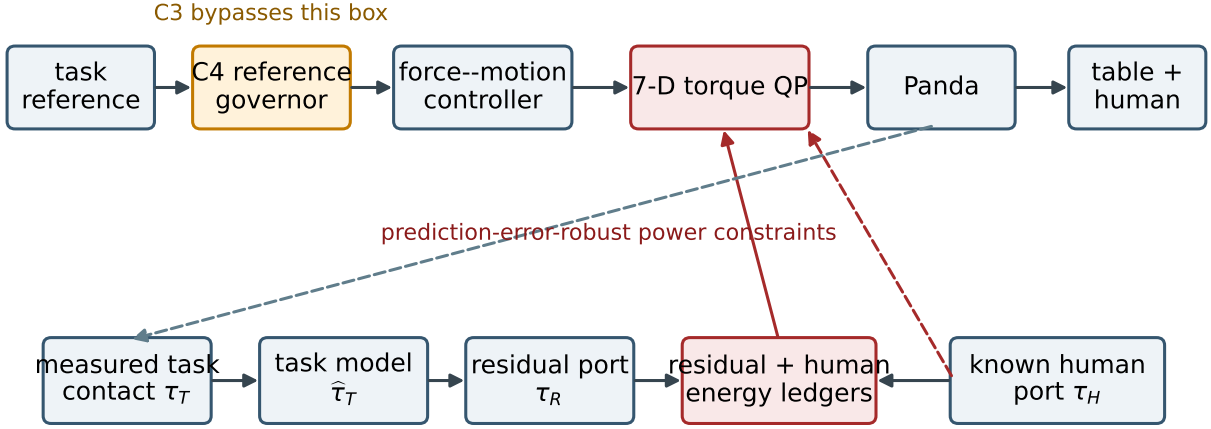


Figure 2: Controller structure. The QP receives separate residual and human accounts. C4 alone modifies the reference upstream of the unchanged C3 QP.

Selective-passivation filter

At each sample, contact and human wrenches are mapped to generalized forces. The predicted task-contact contribution is removed before accounting:

$$\tau_R = \tau_T + \tau_H - \hat{\tau}_T, \quad p_i = \tau_i^\top \dot{\mathbf{q}}, \quad i \in \{R, H\}. \quad (1)$$

Positive power enters the robot. Each ledger uses realized held-sample power,

$$E_{i,k+1} = \min\left(E_{i,\max}, E_{i,k} + \Delta t p_{i,k}^{\text{actual}}\right), \quad (2)$$

and is not lower-clamped, so a violation remains visible in the log. The one-step arm-velocity model is affine in the complete applied torque,

$$\dot{\mathbf{q}}_{k+1}^{\text{pred}} = \mathbf{b}_k + \mathbf{B}_k \boldsymbol{\tau}, \quad \underline{p}_i(\boldsymbol{\tau}) = \tau_i^\top \dot{\mathbf{q}}_{k+1}^{\text{pred}} - \|\tau_i\| \bar{e}_q. \quad (3)$$

The seven-variable QP stays close to the nominal torque and predicted tool twist while enforcing the residual and, for C3/C4, human-ledger conditions

$$\underline{p}_i \geq \max \left[\frac{E_{i,\text{safe}} - E_i}{\Delta t}, -k_{\text{cbf}}(E_i - E_{i,\text{safe}}) \right], \quad \underline{p}_H \geq -P_H^{\max}, \quad (4)$$

together with torque, torque-rate, joint-position, and joint-velocity bounds. The implemented values are $k_{\text{cbf}} = 8 \text{ s}^{-1}$, $P_H^{\max} = 0.18 \text{ W}$, $E_H(0) = 0.055 \text{ J}$, $E_{H,\min} = 0.005 \text{ J}$, and $\bar{e}_q = 0.003 \text{ rad/s}$. The architecture is summarized in Fig. 2.

2 Proof-of-concept results

2.1 Nominal contact and the raw contact signal

Without human input, both reference motions met the nominal targets (Table 2). The compliant-band contact metric was 100% and no joint-limit violation occurred. Both this metric and the raw MuJoCo contact count and force are retained at the 1-ms solver scale. The final pad keeps the nominal raw constraint continuously active.

Table 2: Nominal C0 validation during established motion.

Motion	force RMSE [N]	planar RMSE [mm]	orientation RMSE [deg]	contact [%]	joint violation
straight slide	0.302	3.85	0.96	100	0
bounded wipe	0.378	2.91	0.67	100	0

The contact retuning replaced the single sphere with a finite four-element pad. A 100-ms critically damped contact with low activation-boundary impedance prevents constraint switching, a 600-ms cosine ramp removes the tangential velocity step, and a 70-ms causal filter attenuates the small remaining force variation. Figure 3 shows the unfiltered result rather than only the controller signal. Raw contact occupancy rose from 48.3% to 100%, the raw force peak fell from 12.64 to 5.54 N, and raw-force standard deviation fell from 5.31 to 0.10 N. Normal-velocity standard deviation fell from 1.49 to 0.175 mm/s. The raw signal is therefore already smooth before filtering.

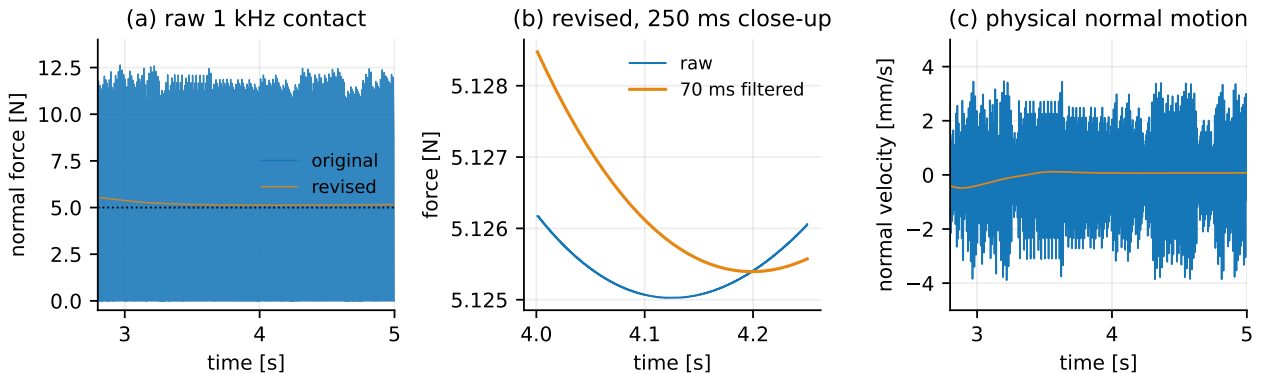


Figure 3: Raw contact evidence. (a) The finite pad and soft activation boundary remove the unfiltered 1-kHz pulse train. (b) The causal filter makes only a small additional change. (c) The revised physical normal velocity also has less variation.

2.2 Model-aware accounting separates task work and mismatch

In the accurate no-human run, completing the 120-mm slide removed 0.215 J from C1’s whole-port ledger, while C3’s residual ledger stayed at 0.300 J (Fig. 4a). Both tracked similarly over this finite stroke, but only C1 spent its budget on expected table friction. With a deliberately incorrect friction/normal-force predictor, the residual ledger instead fell by 0.103 J while the human ledger remained exactly at 0.055 J (Fig. 4b). Model error is therefore exposed without being mislabelled as human work.

Whole-port accounting also becomes unreliable when its usable budget is small. At 0.04, 0.10, and 0.28 J, C1 required 2,077, 1,269, and 1,535 bounded fallback samples, respectively. These intervals are explicitly uncertified. At 0.58 J it required no fallback, but allowed the same 0.263 J human output as C0. This sweep is a limitation of C1, not a successful C1 certificate.

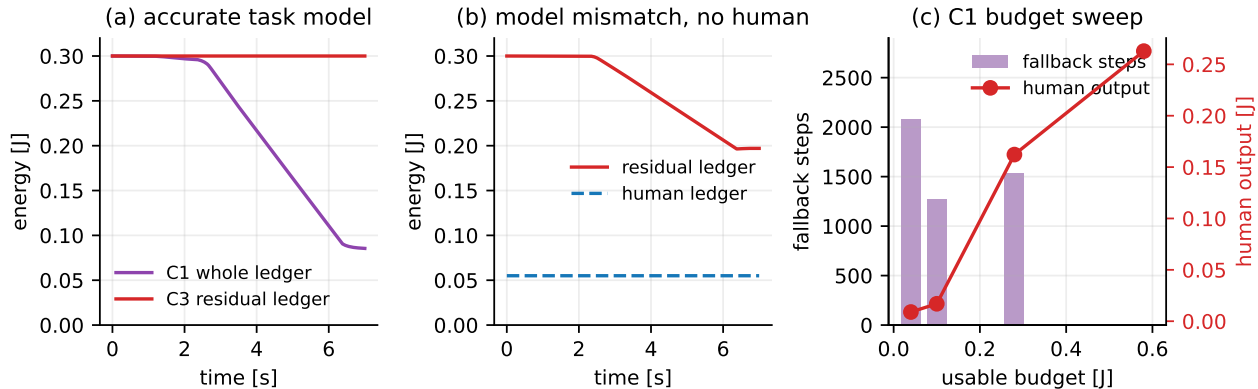


Figure 4: Why task-aware accounting is needed. (a) Whole-port C1 spends energy on ordinary task friction; C3 does not. (b) Predictor mismatch appears only in the residual ledger. (c) Small C1 budgets cause fallback, whereas a large feasible budget ceases to limit human output.

2.3 The dedicated human port limits transferred energy

During oblique end-effector blocking, C0 and residual-only C2 each delivered 0.263 J through the human port. C3 reduced this to 0.049 J, close to the 0.050-J usable human-ledger budget, and limited peak output power to 0.148 W (Fig. 5). It did so by slowing and briefly retreating. During the complete sliding phase, planar RMSE rose to 15.7 mm, force RMSE to 2.87 N, and the maintained-contact fraction fell to 89.6%. These values expose the intended safety–task tradeoff rather than suggesting that energy limiting is free.

The forearm experiment tests whether kinematic redundancy can preserve more of the task while the same human ledger is active. C0 delivered 0.158 J. C3 admitted 0.010 J of helping input and limited gross output to 0.060 J, for a net transfer of -0.049 J, while maintaining contact and 4.26-mm planar RMSE (Fig. 6). Here the arm could reconfigure without the large task loss seen for direct tool blocking. Under release/recovery, C3 restored full contact and achieved 1.68-mm planar RMSE over the final 1.5 s.

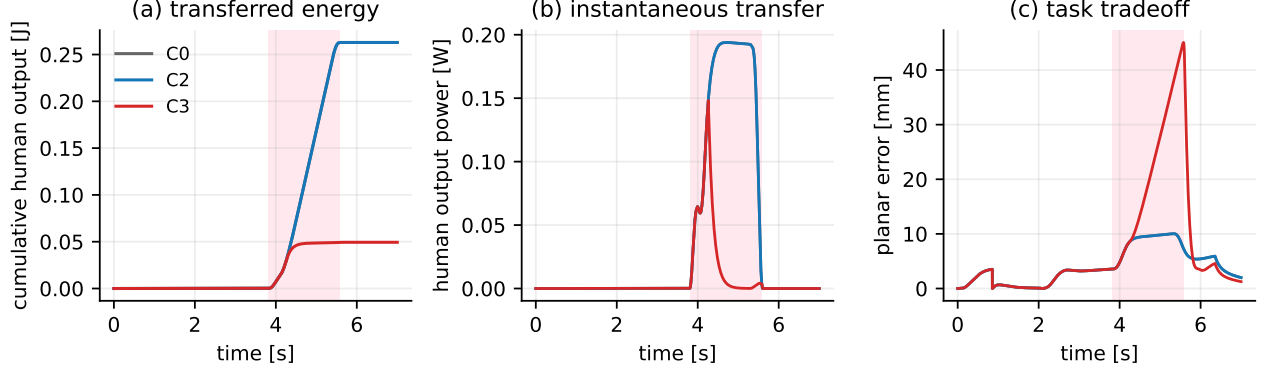


Figure 5: End-effector blocking. The pink region is the human-interaction phase. C3 limits transferred energy and power; C0 and C2 do not have the dedicated human constraint. The increased tracking error is shown at right.

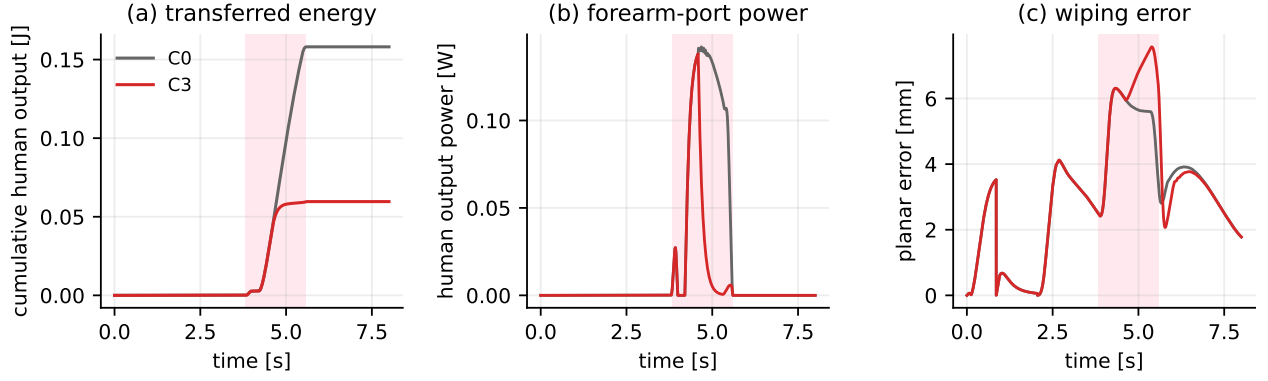


Figure 6: Forearm interaction during wiping. C3 limits net human extraction while using the Panda’s redundancy to keep contact and preserve the wiping path.

2.4 C4 improves recovery without changing the energy result

The C3 QP can slow the robot while the original path continues to advance. C4 smoothly stops that upstream reference while the protected constraint is active and resumes it after release. Human output was effectively unchanged: 0.04947 J for C3 and 0.04946 J for C4. The governor instead reduced complete-run planar RMSE from 12.1 to 6.04 mm and raised maintained contact to 99.7%. Its maximum progress lag was 44.8 mm and returned to zero after release (Fig. 7). The safety effect therefore belongs to the QP; the governor improves how the task reference responds to that intervention.

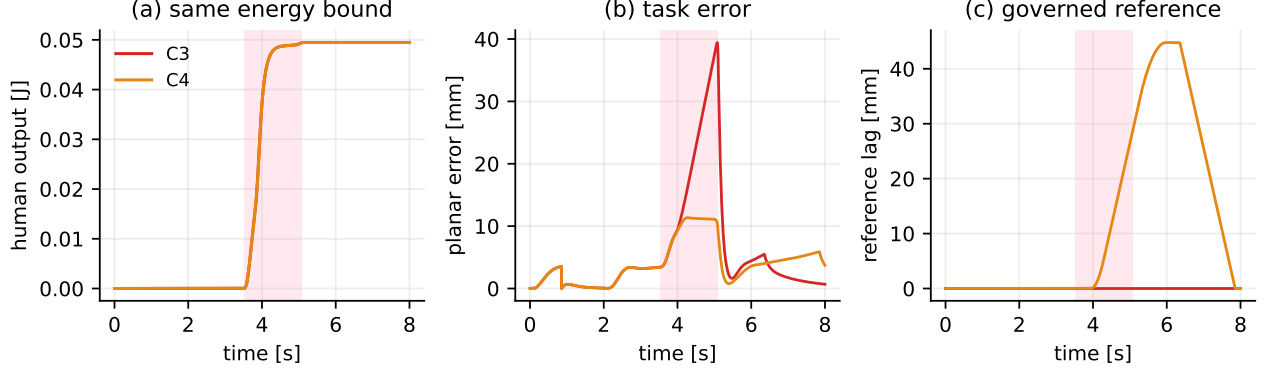


Figure 7: C3 without the governor and C4 with it. Both deliver essentially the same bounded human-port energy. C4 temporarily pauses the reference and substantially reduces tracking error during interaction and recovery.

3 Scope and conclusion

The frozen one-step velocity-error bound was selected after separate calibration: the observed calibration maximum was 0.002180 rad/s, compared with the fixed 0.003 rad/s evaluation bound. The largest C3 evaluation value was 0.002292 rad/s. Across the focused C3 cases there were zero prediction-bound violations, protected-energy-floor violations, and QP fallbacks. In contrast, C1 produced 228 prediction-bound violations across the evaluation set while operating in its fallback regime; those samples are not presented as certified behavior.

The work audit also distinguishes the controller ledger from continuous physical work. The human held-sample versus trapezoidal work discrepancy stayed below 8.5×10^{-5} J in the focused C3 interactions. The largest absolute task-contact discrepancy was 0.00649 J. These differences are reported rather than absorbed into the certificate.

The Panda study reproduces the planar PoC’s central result at a spatial, seven-joint level: predicted task work can be exempted, model mismatch can be retained in a residual account, and a dedicated human account can directly limit energy transfer. It additionally shows that interaction location and redundancy determine the resulting task cost, and that reference governing improves recovery without replacing the passivity filter. The evidence remains conditional on known human wrench and location, the tested predictor, one-step model accuracy, and numerical QP feasibility. Physical sensing, continuous-time guarantees, and broader safety validation remain future work.